

OSAKA, Japan

WMO: 477720

Lat: 34.6822N

Lon: 135.5192E

Elev: 83

StdP: 100.33

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	0.7	1.5	-8.7	1.8	4.5	-7.2	2.1	5.1	10.2	6.1	9.1	5.6	2.9	280	0.394

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.0	34.5	25.0	33.6	24.7	32.5	24.5	26.2	31.6	25.7	31.0	25.4	30.5	3.9	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.8	20.0	28.8	24.3	19.4	28.7	23.8	18.9	28.6	81.8	31.6	80.1	31.1	78.4	30.6	28.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.2	7.0	6.1	-1.1	36.7	1.2	1.0	-1.9	37.3	-2.6	37.9	-3.3	38.4	-4.1	39.1		
			-3.4	27.0	1.4	0.4	-4.4	27.3	-5.2	27.5	-6.0	27.8	-7.0	28.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.4	6.1	6.6	9.9	15.3	20.4	23.9	28.2	29.6	25.6	19.9	13.9	8.6
	DBStd	8.47	2.05	2.63	3.00	3.22	2.41	2.21	2.30	1.89	2.78	2.75	2.92	2.60
	HDD10.0	324	122	99	40	2	0	0	0	0	0	2	58	
	HDD18.3	1528	379	327	262	99	9	0	0	0	15	136	301	
	CDD10.0	3022	1	6	36	162	322	418	563	607	467	307	119	16
	CDD18.3	1183	0	0	0	8	72	168	305	348	217	63	2	0
	CDH23.3	11181	0	0	0	24	297	1034	3416	4331	1847	229	2	0
	CDH26.7	4381	0	0	0	1	35	239	1412	2064	603	26	0	0
Wind	WSAvg	2.7	2.9	2.7	2.8	2.8	2.7	2.7	2.8	2.9	2.6	2.4	2.3	2.7
Precipitation	PrecAvg	1334	46	61	102	120	141	194	161	104	163	118	71	45
	PrecMax	1743	121	196	175	239	361	405	447	341	486	430	171	104
	PrecMin	747	0	10	15	25	25	59	31	7	30	24	4	3
	PrecStd	230	28	36	37	47	74	81	96	75	97	79	41	27
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.1	17.6	20.8	25.8	29.4	32.4	35.6	36.2	33.8	29.3	23.2	17.4
		MCWB	9.9	12.4	13.3	16.8	18.3	23.4	25.2	25.2	24.4	21.7	16.4	12.9
	2%	DB	12.1	14.7	18.6	23.7	27.6	30.4	34.1	35.0	32.5	27.1	21.4	15.5
		MCWB	8.0	9.9	12.2	15.4	17.8	22.3	24.9	25.1	23.8	20.4	15.8	11.4
	5%	DB	10.9	12.6	17.0	22.2	26.2	29.2	33.1	34.1	31.3	25.8	20.0	14.2
		MCWB	7.1	8.3	11.3	14.8	17.5	21.4	24.7	24.8	23.3	19.2	15.1	10.1
	10%	DB	9.9	11.2	15.3	20.7	25.0	28.0	32.1	33.2	30.0	24.5	18.7	13.0
		MCWB	6.0	7.3	10.5	14.2	17.0	20.9	24.4	24.8	22.7	18.3	14.1	9.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.6	13.6	15.7	19.9	22.2	25.0	26.6	26.9	25.8	23.5	18.8	14.7
		MCDB	13.4	16.2	18.5	22.9	25.1	30.1	32.5	32.3	30.7	27.0	20.3	16.2
	2%	WB	9.1	11.1	13.8	17.8	20.8	24.2	26.0	26.4	25.1	21.8	17.2	12.3
		MCDB	11.1	13.5	16.8	20.8	24.0	28.3	31.4	31.8	29.8	25.6	19.9	14.7
	5%	WB	7.6	9.2	12.4	16.5	19.6	23.5	25.6	26.0	24.6	20.6	16.0	10.7
		MCDB	10.3	11.7	15.9	20.4	23.3	27.2	30.7	31.4	29.2	24.3	18.9	13.6
	10%	WB	6.4	7.7	11.0	15.3	18.6	22.6	25.2	25.6	24.1	19.3	14.8	9.4
		MCDB	9.4	10.8	14.7	19.3	22.8	26.2	30.1	31.0	28.5	23.1	18.1	12.4
Mean Daily Temperature Range	5% DB	MDBR	5.9	6.5	7.4	8.2	8.0	6.9	6.4	7.0	6.9	7.0	6.8	6.2
		MCDBR	7.3	8.8	9.7	10.0	9.6	8.4	7.4	7.8	8.0	8.2	7.9	7.6
		MCWBR	5.0	5.8	6.2	5.2	4.2	3.4	2.4	2.6	3.0	3.9	4.7	4.9
	5% WB	MCDBR	6.6	7.4	8.3	7.9	7.3	6.4	6.8	6.8	6.9	6.9	6.7	6.8
		MCWBR	5.5	6.2	6.7	5.4	4.9	2.9	2.5	2.5	3.0	4.4	5.2	5.5
Clear-Sky Solar Irradiance	taub	0.407	0.449	0.512	0.544	0.549	0.569	0.571	0.538	0.493	0.452	0.423	0.402	
	taud	2.126	2.028	1.887	1.860	1.913	1.935	1.980	2.059	2.155	2.184	2.169	2.155	
	Ebn at Noon	766	775	760	758	760	743	738	756	771	767	745	743	
	Edh at Noon	122	150	187	202	194	189	180	164	142	127	116	112	
All-Sky Solar Radiation	RadAvg	2.22	2.86	3.87	4.73	5.20	4.69	4.98	5.10	4.02	3.24	2.45	2.02	
	RadStd	0.16	0.23	0.30	0.41	0.54	0.44	0.65	0.45	0.34	0.32	0.20	0.13	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (33 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page